Yudi Ruan

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Portfolio Website: ruanyudi.github.io

EDUCATION

Chongqing Jiaotong University Chongqing, China

Bachelor's Degree in *Artificial Intelligence* 09/2022-07/2026

- GPA: 4.15/5.0 | Class Rank: **1**/65

SELECTED PUBLICATIONS

Full publication list available on my Google Scholar or Portfolio Website

- **Yudi Ruan**, Hao Ma, Weikai Li, Xiao Wang (2025). Low-light image enhancement using dual cross attention. *Engineering Applications of Artificial Intelligence.*
- **Yudi Ruan**, Di Wang, Yijing Yuan, Shixin Jiang, Xianyi Yang (2025). SKPNet: snake KAN perceive bridge cracks through semantic segmentation. *Intelligence & Robotics*.
- Huaiqu Feng, **Yudi Ruan**, et al. (2025). nUGV-1UAV robot swarms: low-altitude remote sensing-based decentralized planning framework in-field environments. *ISPRS JP&RS.*
- Huaiqu Feng, Te Xi, **Yudi Ruan**, et al. (2025). Topological organization for hybrid rice growth stages Phenotype based on Contrastive clustering. *Expert Systems with Applications*.
- Weikai Li, Hongfeng Wei, Yanlai Wu, Jie Yang, **Yudi Ruan**, et al. (2024). TIDE: Test-Time Few-Shot Object Detection. *IEEE Transactions on Systems, Man and Cybernetics: Systems*.
- Jie Yang, Xiaowen Xu, Mingxiang Sun, **Yudi Ruan**, et al. (2024). Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data. *Cerebral Cortex*.
- Sifu Zeng, Jie Yang, Wang Luo, **Yudi Ruan** (2023). Few-shot segmentation with duplex network and attention augmented module. *Frontiers in Neurorobotics*.

RESEARCH EXPERIENCES

Autonomous nUGV–nUAV Swarm for Precision Agriculture 08/2024-Now

State Key Laboratory–Supported Project | Core Developer | Advised by Dr. Huaiqu Feng

- Designed an intelligent hybrid-rice purification pipeline that fuses expert knowledge with a self-supervised t-SimCLR clustering framework for accurate in-field detection of undesired varieties.
- Engineered the ECS-LRSMP cooperative UGV–UAV motion-planning framework combining Voronoi-based charging deployment, combinatorial optimization, and evolutionary algorithms for efficient, energy-constrained field exploration.
- Implemented a decentralized nUGV–1UAV impurity-removal system utilizing UAV mapping, topology-guided hybrid-A* planning, and multi-UGV space–time optimization to achieve scalable autonomous operations in large paddy fields.

Controllable Video Generation Platform 05/2024-Now

Project Lead & Core Developer

- Developed a video generation framework based on an optimized Diffusion Transformer (DiT) architecture.
- Enabled conditional denoising and synthesis through text, speech, and image control signals, supporting fine-grained video generation.
- Collected large-scale web videos via automated crawlers and built a data cleaning and semantic labeling pipeline to enhance data quality.
- Optimized the inference process, achieving real-time generation at 40 FPS on a single NVIDIA A10 GPU.

Medical 3D Multi-Modal Imaging System 05/2024-07/2025

Project Lead & Core Developer | Advised by Prof. Weikai Li

- Developed a unified 3D medical imaging framework for multi-modal representation learning, segmentation, and visualization.
- Proposed a VAE – Diffusion Transformer architecture to fuse and reconstruct incomplete 3D modalities, enhancing data efficiency and generalization.
- Optimized the nnUNet-based segmentation pipeline, achieving a Dice score of 89.3 on multi-organ datasets.
- Built an interactive VTK-based visualization module and deployed scalable containerized inference services using Docker and Kubernetes.

OmniVision: Integrated Image Enhancement System 02/2024-07/2025

National Innovation Project | Lead Developer | Advised by Prof. Weikai Li

- Collected and preprocessed large-scale traffic image datasets to support model training and evaluation.
- Converted state-of-the-art vision models into ONNX format and deployed them on a unified inference platform, enabling cross-device compatibility and real-time acceleration.
- Designed the ECAFormer model, delivering nearly a 3% PSNR improvement over the strongest baseline.

Multi-Sensor Fusion Navigation System for Urban Rail 05/2023-05/2024

National Innovation Project | Project Lead & Core Developer | Advised by Prof. Wei Tan

- Deployed deep learning models on NVIDIA Jetson using TensorRT with GPU and DLA accelerators for real-time inference.
- Parallelized custom algorithms with CUDA to achieve low-latency computation and improve efficiency.
- Built multi-process, multi-machine communication pipeline via ROS for real-time data transmission.
- Achieved odometry accuracy within 1% through LiDAR–Visual–IMU–GNSS sensor fusion.

INTERNSHIPS

Hedra Inc. CA, U.S.

Deep Learning Intern (Remote) – Real-Time Interactive Avatar 10/2024-08/2025

- Led real-time optimization of a video generation model from MVP design to production deployment (see: https://livatar-1.github.io/Livatar-1).
- Accelerated model performance by optimizing computation steps with Torch Dynamo and implementing feature caching without sacrificing accuracy.
- Containerized and deployed the system with Docker; integrated AWS ECR, EKS, and SQS to enable auto-scaling and queue-based task monitoring.

iFLYTEK Chongqing, China

Software Development Intern (On-site) 06/2025-07/2025

- Built and deployed key modules of an in-vehicle smart system using TensorFlow, STM32, and Flask, including voice recognition and environmental sensing (temperature & humidity).

HONORS

- National Endeavor Scholarship (1%) 2024
- University Merit Student (3%) 2024
- National Scholarship for Outstanding Undergraduates (1%) 2023
- University Merit Student (3%) 2023
- University Academic Excellence Awardee (3%) 2023
- University Innovation Pioneer Awardee (3%) 2023

AWARDS IN ACADEMIC COMPETITIONS

- National 1st Prize, China Business Elite Challenge (5%) 10/2024
- National 1st Prize, RAICOM Robotics Developer Competition (3%) 08/2024
- National 2nd Prize, China Robotics & AI Competition (10%) 08/2024
- Regional Silver Award, "Internet+" Innovation Competition (30%) 08/2024
- National 3rd Prize, Smart Transportation Innovation Competition (30%) 07/2024
- Regional 2nd Prize, "Datang Cup" ICT Innovation Competition (15%) 05/2024
- H Award, Mathematical Contest in Modeling, COMAP (USA) (30%) 01/2024
- Regional 1st Prize, Math Modeling Contest (8%) 11/2023
- Regional 2nd Prize, Intelligent Car Competition (iFLYTEK Track) (10%) 09/2023
- Regional 2nd Prize, Intelligent Car Competition (Baidu Track) (10%) 08/2023
- Regional 3rd Prize, "China Software Cup" (10%) 07/2023
- Regional 2nd Prize, College Computing Competition (10%) 05/2023
- 1st Prize, International Artificial Intelligence Fair (3%) 08/2020

ADDITIONAL INFORMATION

- **Programming:** C++ (6+ yrs), Python (3+ yrs), Java (1+ yr).
- **Databases:** MySQL (2+ yrs), SQLite (2+ yrs).
- **Libraries & Frameworks:** NumPy (3+ yrs), Pandas (3+ yrs), CUDA (3+ yrs), PyTorch (2+ yrs).
- **Tools & Platforms:** Docker (2+ yrs), Kubernetes (2+ yrs).
- **Languages:** Chinese (Native), English (Proficient).
- **Hobbies:** Alpine climbing, Music, Traveling.